

Overview of Attention is All You Need

Subject: Attention is All You Need

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

1.

"Attention Is All You Need" is a 2017 research paper in machine learning authored by eight scientists and engineers working at Google. The paper introduced a new deep learning architecture known as the transformer, based on the attention mechanism proposed in 2014 by Bahdanau et al. The transformer approach it describes has become the main architecture of a wide variety of artificial intelligence, including large language models. At the time, the focus of the research was on improving Seq2seq techniques for machine translation, but the authors go further in the paper, foreseeing the technique's potential for other tasks like question answering and what is now known as multimodal generative AI. Some early examples that the team tried their Transformer architecture on included English-to-German translation, generating Wikipedia articles on "The Transformer", and parsing. These convinced the team that the Transformer is a general-purpose language model, and not just good for translation. As of 2025, the paper has been cited more than 173,000 times, placing it among the top ten most-cited papers of the 21st century. After the paper was published by Google, each of the authors left the company to join other companies or to found startups.

2.

While the primary focus of the paper at the time was to improve machine translation, the paper also discussed the use of the architecture on English Constituency Parsing, both with limited and large-sized datasets, achieving a high-score without specific tuning for the task indicating the promising nature of the model for use in a wide-variety of general purpose of seq2seq tasks.

3.

Methods discussed and introduced The paper is best known for introducing the Transformer architecture, which underlies most modern large language models (LLMs). A key reason why the architecture is preferred by most modern LLMs is the parallelizability of the architecture over its predecessors. This ensures that the operations necessary for training can be accelerated on a GPU, allowing both faster training times and models of bigger sizes to be trained. The paper introduced the following mechanisms as part of the development of the transformer architecture.

4.

Multi-head attention In the self-attention mechanism, queries (Q), keys (K), and values (V) are dynamically generated for each input sequence (typically limited by the size of the context window), allowing the model to focus on different parts of the input sequence at different steps. Multi-head attention enhances this process by introducing multiple parallel attention heads. Each attention head learns different linear projections of the Q, K, and V matrices. This allows the model to capture

different aspects of the relationships between words in the sequence simultaneously, rather than focusing on a single aspect. By doing this, multi-head attention ensures that the input embeddings are updated from a more varied and diverse set of perspectives. After the attention outputs from all heads are calculated, they are concatenated and passed through a final linear transformation to generate the output.

5.

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) \quad \{\displaystyle PE_{\left(\left\{\rm\{pos\}\}, 2i\right)\right)} = \sin\left(\left\{\rm\{pos\}\} / \left\{10000\right\}^{\left\{2i/d_{\rm\{model\}}\right\}\right)}\right\}$$

6.

Predecessors For many years, sequence modelling and generation was done by using plain recurrent neural networks (RNNs). A well-cited early example was the Elman network (1990). In theory, the information from one token can propagate arbitrarily far down the sequence, but in practice the vanishing-gradient problem leaves the model's state at the end of a long sentence without precise, extractable information about preceding tokens. A key breakthrough was LSTM (originally described in a 1995 technical report and formally published in 1997), an RNN that introduced gating mechanisms to mitigate the vanishing gradient problem, allowing efficient learning of long-sequence modelling. One key architectural element was the use of multiplicative gating units, in which the outputs of some neurons modulate the outputs of others. These multiplicative units are conceptually distinct from the additive attention mechanism later introduced for sequence-to-sequence models. Neural networks using multiplicative units were later called sigma-pi networks or higher-order networks. LSTM became the standard architecture for long sequence modelling until the 2017 publication of transformers. However, LSTM still used sequential processing, like most other RNNs. Specifically, RNNs operate one token at a time from first to last; they cannot operate in parallel over all tokens in a sequence. Modern transformers overcome this problem, but unlike RNNs, they require computation time that is quadratic in the size of the context window. The linearly scaling fast weight controller (1992) learns to compute a weight matrix for further processing depending on the input. One of its two networks has "fast weights" or "dynamic links" (1981). A slow neural network learns by gradient descent to generate keys and values for computing the weight changes of the fast neural network which computes answers to queries. This was later shown to be equivalent to the unnormalized linear transformer.

7.

$$Attention(Q, K, V) := softmax\left(\frac{Q \times K^T}{d_k}\right) \times V \quad \{\displaystyle \left\{\rm\{Attention\}\right\}(Q, K, V) := \left\{\rm\{softmax\}\right\}\left(\frac{Q \times K^T}{\sqrt{d_k}}\right) \times V\}$$

8.

Positional encoding Since the Transformer does not rely on recurrence or convolution of the text in order to perform encoding and decoding, the paper relied on the use of sine and cosine wave functions to encode the position of the token into the embedding. The methods introduced in the paper are discussed below:

9.

Scaled dot-product attention and self-attention The use of the scaled dot-product attention and self-attention mechanism instead of a recurrent neural network or long short-term memory (which rely on recurrence instead) allows for better performance as described in the following paragraph. The paper described the scaled dot-product attention as follows:

10.

Seq2seq models with attention (including self-attention) still suffered from the same issue with recurrent networks, which is that they are hard to parallelize, which prevented them from being accelerated on GPUs. In 2016, decomposable attention applied a self-attention mechanism to feedforward networks, which are easy to parallelize, and achieved SOTA result in textual entailment with an order of magnitude fewer parameters than LSTMs. One of its authors, Jakob Uszkoreit, suspected that attention without recurrence would be sufficient for language translation, thus the title "attention is all you need". That hypothesis was against conventional wisdom at the time, and even his father Hans Uszkoreit, a well-known computational linguist, was skeptical. In the same year, self-attention (called intra-attention or intra-sentence attention) was proposed for LSTMs. On 2017-06-12, the original (100M-parameter) encoder–decoder transformer model was published in the "Attention is all you need" paper. At the time, the focus of the research was on improving seq2seq for machine translation, by removing its recurrence to process all tokens in parallel, but preserving its dot-product attention mechanism to keep its text processing performance. This led to the introduction of a multi-head attention model that was easier to parallelize due to the use of independent heads and the lack of recurrence. Its parallelizability was an important factor to its widespread use in large neural networks.